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Recommendations for Cassandra



The configurations listed below result from the experience Feedzai has running this system. Nonetheless, it is important to review the configurations in light of the workload requirements, and adjusted accordingly.

Cassandra Configuration

The `cassandra.yaml` file is the main configuration file for Cassandra.

Property	Recommended Value	Description
cluster_name	Feedzai Cassandra Cluster	This setting prevents nodes in one logical cluster from joining another. All nodes in a cluster must have the same value.
num_tokens	256	Set this property for virtual node token architecture. Determines the number of token ranges to assign to this [vnode].
endpoint_snitch	GossipingPropertyFileSnitch (see note below)	The implementation used informs Cassandra about the network topology to route requests efficiently, and allows Cassandra to spread replicas around your cluster to avoid correlated failures.
concurrent_reads	8 * #cores	
concurrent_writes	8 * #cores	
trickle_fsync	true	Causes fsync to force the operating system to flush the dirty buffers at the set interval <code>trickle_fsync_interval_in_kb</code> , to prevent sudden flushes from impacting read latencies.
trickle_fsync_interval_in_kb	4096	The size of the fsync in kilobytes.
concurrent_compactors	#cores	The number of concurrent compaction processes allowed to run simultaneously on a node.
compaction_throughput_mb_per_sec	0	Throttles compaction to the specified Mb/second across the instance. Setting the value to 0 disables compaction throttling.
disk_failure_policy	stop	On disk failures, shuts down gossip and client transports, leaving the node effectively dead, but it still can be inspected with JMX.
commit_failure_policy	stop	On commit disk failures, shuts down gossip and client transports, leaving the node effectively dead, but it still can be inspected with JMX.
commitlog_sync	periodic	Ack writes immediately and sync the CommitLog every <code>commitlog_sync_period_in_ms</code>
commitlog_sync_period_in_ms	10000	Frequency of the CommitLog sync when in <code>periodic</code> mode.
auto_bootstrap	true	It causes new (non-seed) nodes migrate the right data to themselves automatically.



Endpoint Snitch

With the `GossipingPropertyFileSnitch` the rack and datacenter for the local node must be defined in `cassandra-rackdc.properties`.

If you are installing Cassandra on AWS, the recommendation is to use [Ec2Snitch](#). This will rely on the instances and the availability zones they are done to define a better and more reliable topology.

If you are installing Cassandra on GCP, there is a [GoogleCloudSnitch](#) available.



Make sure `ntp` is installed, and all Cassandra servers are in-sync.

Java

Java Native Access (JNA)

Using JNA is one way to improve Cassandra memory usage, and, hence, performance. If JNA is not installed and enabled, Cassandra will not boot unless `cassandra.boot_without_jna` is set to true.

JVM Opts

It is advisable that the `-Xms` and `-Xmx` flags are set to the same value. Optionally, `-Xmn` could be set as well, according to the resources and workload.

Make sure you define the most appropriate GC algorithm according to the workload and the heap size.